

Riding Wavelets: Data-Mining Fundamental Knowledge in Financial Time Series - MVA 2025/26

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1 Introduction

Financial datasets record stock prices of multiple assets over time and constitute a primary source for studying financial markets and their behavior. The data are extreme and highly non-stationary across all scales, from minutes to days. The features to be extracted are therefore very difficult to identify and are not publicly disclosed by hedge funds. Non-stationarity makes dictionary learning unstable, so we instead rely on analyses based on spectral features. According to the paper, applying PCA in the wavelet space successfully captures these features using data from multiple jumps across financial markets.

This report presents our open source implementation <https://github.com/janisaiad/timeseries> of the jump detection and classification methodology proposed by Aubrun et al. [1]. The paper introduces an unsupervised framework based on wavelet coefficients to classify financial price jumps according to their reflexivity, mean-reversion, and trend-alignment properties.

Our implementation is based on code developed from scratch and public findable, accessible, interpretable, reusable data available at [4], and tests the methodology across different countries and sampling frequencies. The jump detection and wavelet transformation code was developed by Janis Aiad, while Jordi Baroja focused on hyperparameter testing and finding correlations between features. Both authors fully understood all aspects of the paper, ran numerical experiments, and drew conclusions jointly.

2 Method

2.1 Jump Detection

Following the paper’s methodology, we compute the log returns for a given stock price time series as $r(t) = \log \left(\frac{P(t+1)-P(t)}{P(t)} \right)$. While the data in the paper is recorded every 1 minute, we also work with 5-minute, hourly, and daily data. The normalization of the data is performed as follows:

$$x(t) = \text{sign} \left(\frac{r(0)}{f(0)\sigma(0)} \right) \frac{r(t)}{f(t)\sigma(t)} \quad \text{Forcing } x(0) > 0 \quad (1)$$

The term $f(t)$ represents the intraday volatility pattern (U-shape), computed as the median of returns at each time of day for a given stock across multiple days, and normalized to have unit quadratic mean. The term $\sigma(t)$ is the exponentially weighted moving standard deviation (EWMA), which computes the standard deviation of past values with exponential decay. We changed this estimator since it produces better results than the one the paper uses. As stated in the paper [1], under the null hypothesis of Gaussian residuals (no-jump hypothesis), $|x(t)|$ converges towards a Gumbel distribution.

Theoretically, jumps are detected when $|x(t)| > 3.95$, corresponding to a 4-sigma deviation. We filter trading hours to exclude the first and last 60 minutes of each trading day (10:30-15:00 window).

2.2 Wavelet Feature Extraction

We implement the wavelet scattering representation $\Phi(x)$. For that, we use a complex wavelet filter $\psi(t)$ whose Fourier transform is real, has 0 average and the real part is even while the imaginary part is odd. Several functions exist with those properties and are well established to be good at feature extraction and can be found in (PyWavelets) [3].

- Primary coefficients:

$$W_j x(t) := x \star \psi_j(t), \quad j \in \{1, \dots, J\}, \quad \text{where } \star \text{ denotes convolution and } \psi_j(t) = \psi(2^{-j}t).$$

This leads to $2J$ coefficients, as we save the real and imaginary part evaluated at $t = 0$.

- Second-order coefficients:

$$W_{j_2} |W_{j_1} x|(t) := |x \star \psi_{j_1}| \star \psi_{j_2}(t), \quad 1 \leq j_1 < j_2 \leq J, \quad \text{where } |\cdot| \text{ denotes the absolute value.}$$

This yields $J(J-1)$ coefficients since we save $\binom{J}{2}$ imaginary and real coefficients evaluated again at 0.

For $J = 6$ scales, this yields 42 features representation so $\Phi(x) \in \mathbb{R}^{42}$. Recall that every normalized jump x has T components, being x_0 the moment of the jump and saving log returns from $x_{-T/2}$ until $x_{T/2}$. In our case $T = 25$.

2.3 Scattering Spectra

We also implement the Scattering Spectra (SS) features as described in Morel et al. [2]:

- $\Phi_1[j]$: Low-moment kurtosis measure, $\Phi_2[j]$: Average volatility at scale 2^j
- $\Phi_3[j, j']$: Multi-scale skewness (complex), $\Phi_4[j_1, j'_1, j_2]$: Multi-scale kurtosis via second-order scattering (complex)

2.4 Kernel PCA and Classification

After we compute the wavelet and scattering spectra representation of each processed jump x , we apply Kernel PCA (RBF kernel, linear kernel) to obtain a low-dimensional embedding. The first principal component D_1 captures volatility asymmetry (reflexivity). We also compute handcrafted features:

- D_2 : Mean-reversion = $x(-1) - x(+1)$
- D_3 : Trend = $x(-1) + x(+1)$
- Asymmetry:

$$A_{\text{jump}} = \frac{A_{>} - A_{<}}{A_{>} + A_{<}}, \quad \text{where } A_{< / >} := \sum_{t_{< / >} > 0} |x(t) - \min_{t_{< / >} > 0} x(t)|$$

2.5 Description of experiments

In the experiments we conduct, we focus on testing whether including scattering spectra features improves classification. We also compare different KPCA methods (RBF and Linear), vary the J value in the wavelet representation, and perform tests across different countries to see if PCA captures the same features. Additionally, we examine the relationship between learned features and hand-crafted features—a comparison not performed in the original paper—using different timescales and plotting feature quantiles to visualize the extracted patterns.

freq	n stocks	total points	span (days)	start	end
5min	80	477302	148	07/07/2025 10:00	12/02/2025 15:40
hourly	80	105700	628	04/11/2024 11:00	12/30/2025 16:00
daily	80	254918	10205	01/21/1998	12/30/2025

Table 1: Poland dataset summary (after preprocessing).

3 Data

We use open source Stooq OHLC data for multiple stock markets and multiple sampling frequencies (5-min, hourly, daily; see Tables 1). Since the original paper uses 1-minute data with a 119-point window and $J = 6$, we adapt to coarser bars with shorter windows and set $J = 3$ for 5-min experiments.

Preprocessing diagnostics: jump-score distribution and tails

The standardized returns exhibit heavy tails; the upper tail of $|x(t)|$ is well captured by a Gumbel fit on a high-quantile range, consistent with the max-stable null model discussed in the paper. We aggregate 12×5 -min bars into hourly and $12 \times$ hourly into daily; since the number of observations (and thus extreme events) decreases sharply at coarser scales, we use lower practical thresholds at coarser frequencies (roughly $\tau \approx 4$ for 5-min, $\tau \approx 3$ for hourly, $\tau \approx 2$ for daily) to retain enough jumps for stable statistics. We note that the 5-min tail shows a mild deviation from the Gumbel fit in the far tail; this could indicate a heavier-than-exponential tail (suggesting a Fréchet/GEV-type behavior) rather than a pure Gumbel regime, but we do not push this interpretation further here. We also note that our volatility estimator was adjusted during development; a deeper extreme-value analysis (e.g., GEV/GPD fits on exceedances) is left for future work.

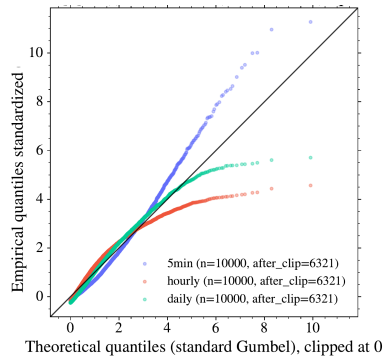


Figure 1: Q-Q plot of $|x(t)|$ against the fitted Gumbel distribution (stacked across 5-min, hourly, daily).

Data conclusion. Overall, the preprocessing checks support the hypotheses needed for the wavelet pipeline: after de-seasonalization and volatility normalization, jump scores have stable heavy-tail behavior compatible with the paper’s extreme-value motivation, and the learned directions exhibit well-behaved empirical distributions. This gives confidence that applying the embedding and interpretability analysis below is meaningful on our datasets.

4 Results

4.1 Hyperparameter test

Ablation: Scattering Spectra features Adding SS features yields very similar D_1 embeddings (high correlation), suggesting that the base wavelet scattering coefficients already capture the dominant reflexivity signal at this resolution.

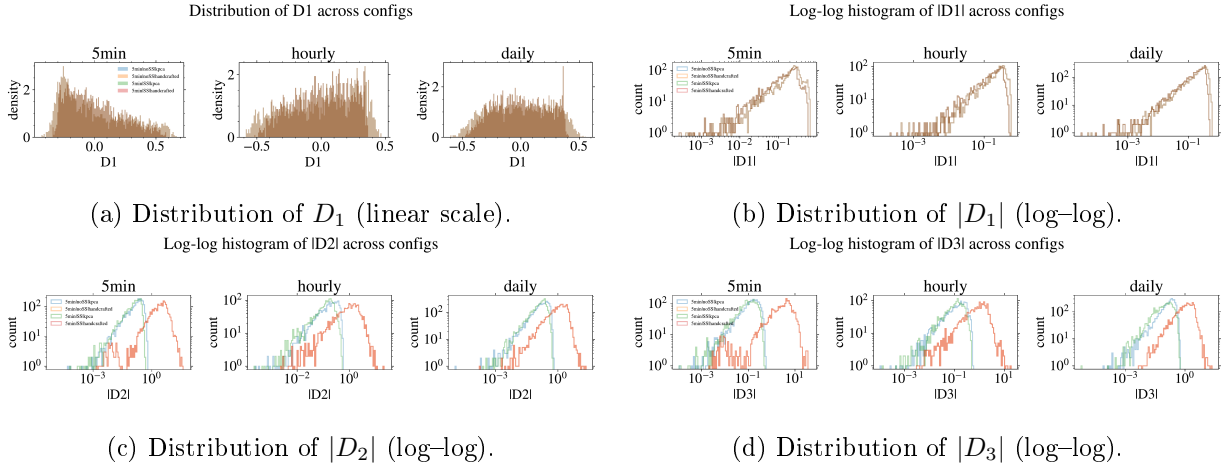


Figure 2: Distribution diagnostics (Poland, 5-min): reflexivity score D_1 and tails of $|D_1|$, $|D_2|$, $|D_3|$.

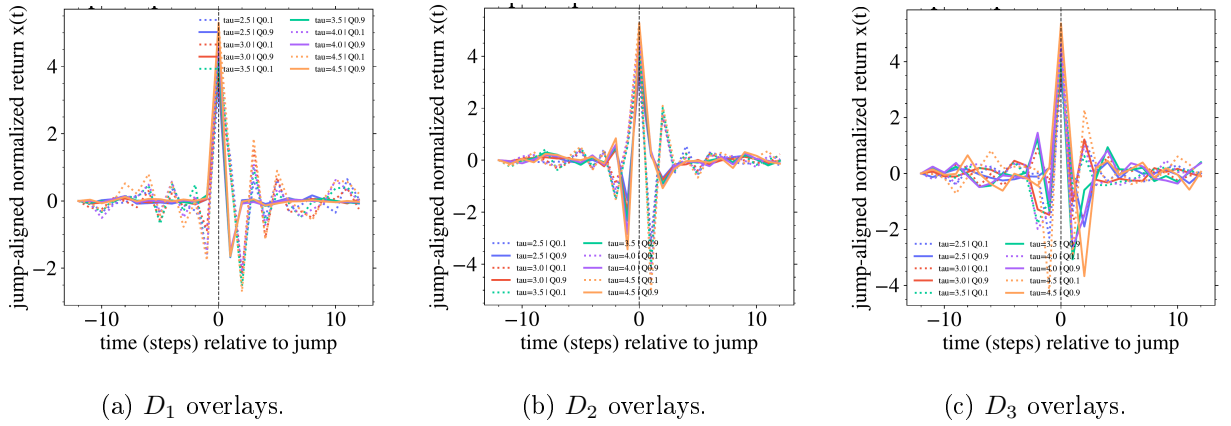


Figure 3: Threshold sweep (5-min): average profiles along D_1 , D_2 , D_3 remain qualitatively stable across thresholds.

Robustness to the jump threshold (5-min) We evaluate the sensitivity of the discovered directions to the jump-detection threshold τ . For 5-min data, we observe a striking stability across thresholds (e.g., $\tau \in [2.5, 4.5]$): D_1 and D_3 profiles are nearly unchanged, while D_2 can degrade slightly at the most extreme thresholds due to a small set of outliers that do not follow the typical mean-reversion pattern. Supplementary overlays for hourly and daily data are provided in 3.

Kernel PCA method: we tested the algorithm using both the RBF method and the linear method. The first one applies a kernel in an infinite-dimensional Hilbert space, while the latter is simply standard PCA on the 42-dimensional wavelet feature space. The results in 7 show that simple PCA performs as well as the more sophisticated RBF method, since it learns the same representation.

Wavelet: after testing the learned directions in 8 and visualizing the top and lowest quantiles in each direction in 9, we decided to keep the most common choice, the Morlet wavelet with bandwidth 1.5 and center frequency 1.0, since all wavelets give similar results but this one produces smoother shapes.

J parameter: by increasing the parameter J , we increase the dimension of the wavelet feature space. Surprisingly, as shown in 4, increasing it up to $J = 9$ does not better capture the properties we are interested in. For instance, the second direction captures mean reversion more clearly with $J = 3$, which should theoretically be the worst-performing setting. We conclude that the learned directions are not very stable with respect to hyperparameter changes, and that each configuration must be empirically evaluated.

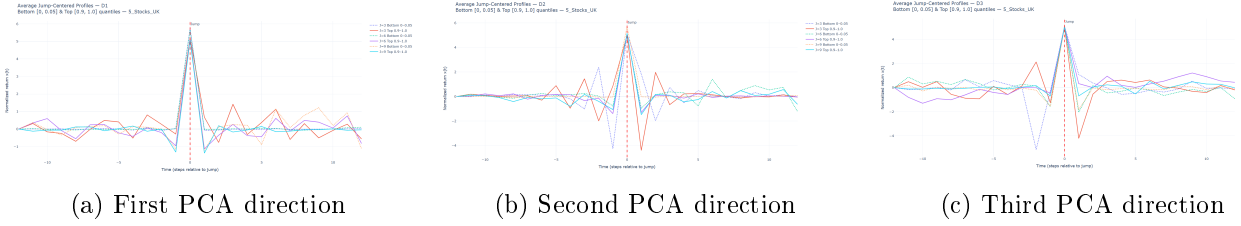


Figure 4: Visualization of first and last quantiles jumps on UK stock data for different values of J .

Different countries: since the data come from several countries, we can test whether the algorithm consistently learns the same features. As shown in 5, the learned directions are quite similar across countries. The paper states that the first direction is a linear combination of the 15 imaginary second-order wavelet coefficients. While this interpretation is conceptually appealing, as it relates to time asymmetry, our results suggest that this direction is not purely that feature.

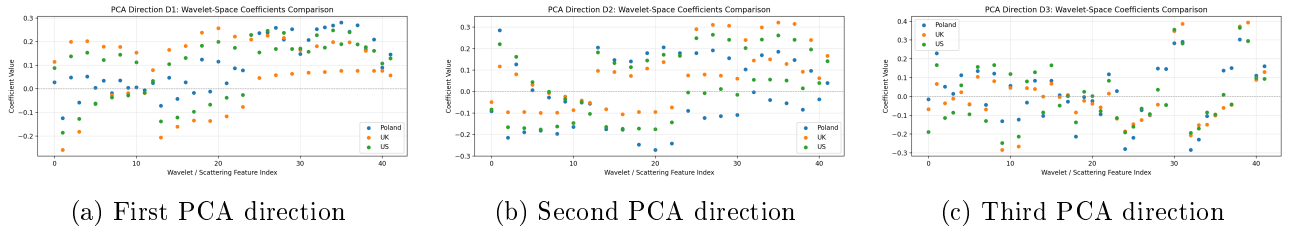


Figure 5: Learned directions for the first 3 PCA direction from data of 3 different countries: Poland, UK and US.

4.2 Relation between learned features and hand crafted features

Here we address the following question: are the handcrafted features relevant to the features learned via PCA on wavelet coefficients? This question is central to the original paper, which claims that well-established financial features can be recovered in an unsupervised manner. We examine the correlations between handcrafted and learned features, as shown in 6. The first PCA direction is the best explained by its corresponding handcrafted feature, exhibiting a strong positive correlation and a non-negligible R^2 value. The third direction also correlates with the trend feature (correlation of 0.32), while the second direction clearly differs from mean reversion. This suggests that unsupervised feature extraction does not necessarily recover the expected financial descriptors. It may also indicate that other latent properties of the data are relevant, that alternative feature representations could be informative, or that the difference arises from using 5-minute data instead of 1-minute data.

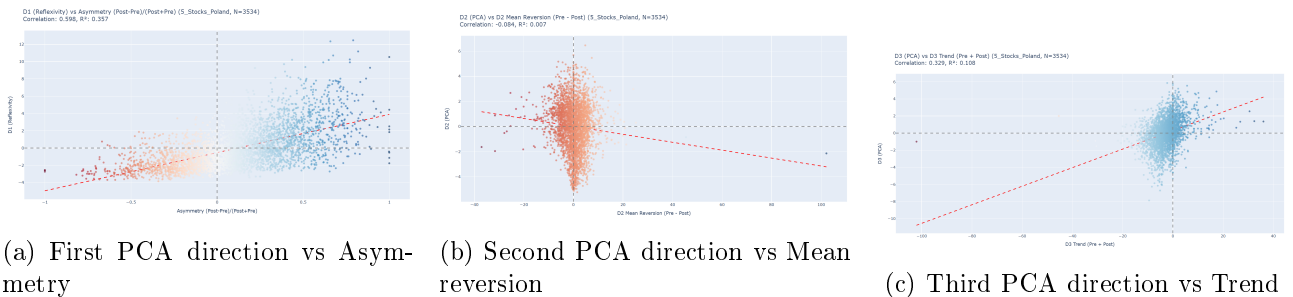


Figure 6: Correlation between learned PCA directions and hand crafted features. Experiment done with 5 minutes Poland stock

References

- [1] Cecilia Aubrun, Rudy Morel, Michael Benzaquen, and Jean-Philippe Bouchaud. Riding wavelets: A method to discover new classes of price jumps, 2024.
- [2] Rudy Morel, Gaspar Rochette, Roberto Leonarduzzi, Jean-Philippe Bouchaud, and Stéphane Mallat. Scale dependencies and self-similarity through wavelet scattering covariance. *arXiv preprint arXiv:2204.10177*, 2022.
- [3] PyWavelets Developers. Pywavelets: A python package for wavelet analysis. <https://pywavelets.readthedocs.io>, 2026. Accessed: 2026-01-04.
- [4] Stooq.com. Stooq historical data. <https://stooq.com/db/h/>, 2026. Accessed: 2026-01-04.

Appendix

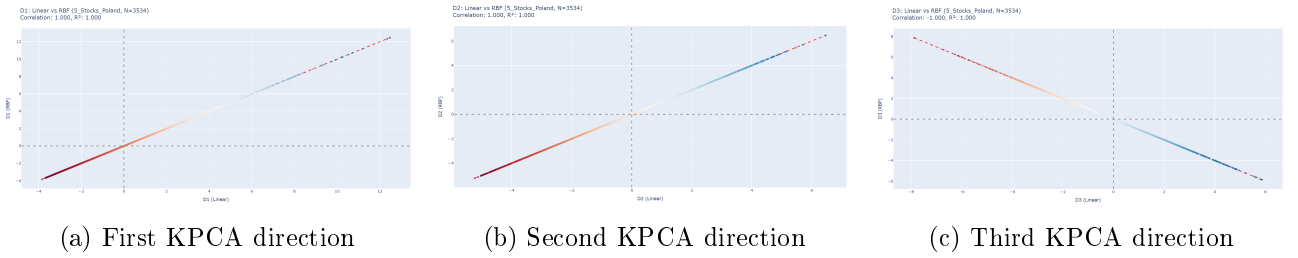


Figure 7: Projected jumps from Polish stocks onto the first three KPCA directions. Shows correlation between the RBF and linear methods.

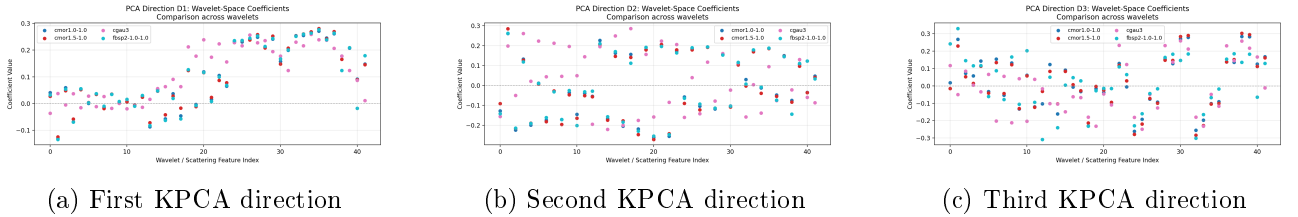


Figure 8: Learned PCA directions in wavelet space on Poland stock data for wavelets: 2 complex Morlet (cmor), 1 complex Gaussian (cgau) and a frequency B spline (fbsp). Each graph represents the direction coefficients for each of the 42 wavelet components.

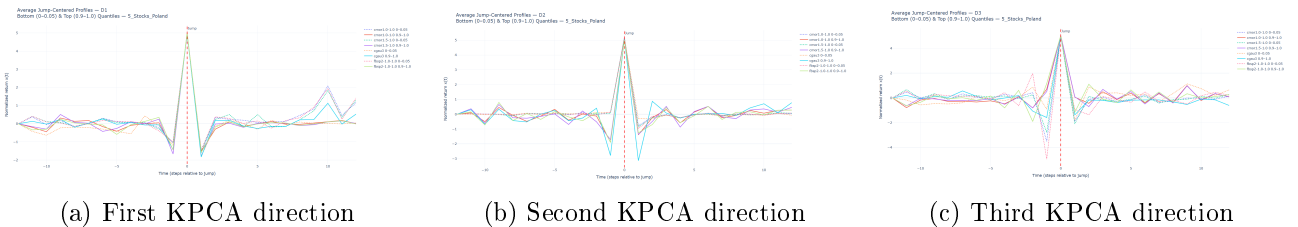


Figure 9: Visualization of first and last quantiles jumps on Poland stock data for wavelets: 2 complex Morlet (cmor), 1 complex Gaussian (cgau) and a frequency B spline (fbsp).

4.3 Threshold sweep overlays across frequencies

This supplementary appendix provides the full set of threshold-sweep overlay plots across sampling frequencies (5-min, hourly, daily) for the three main directions (D_1 , D_2 , D_3). The main report includes the 5-min overlays; the plots below show the complete comparison grid.

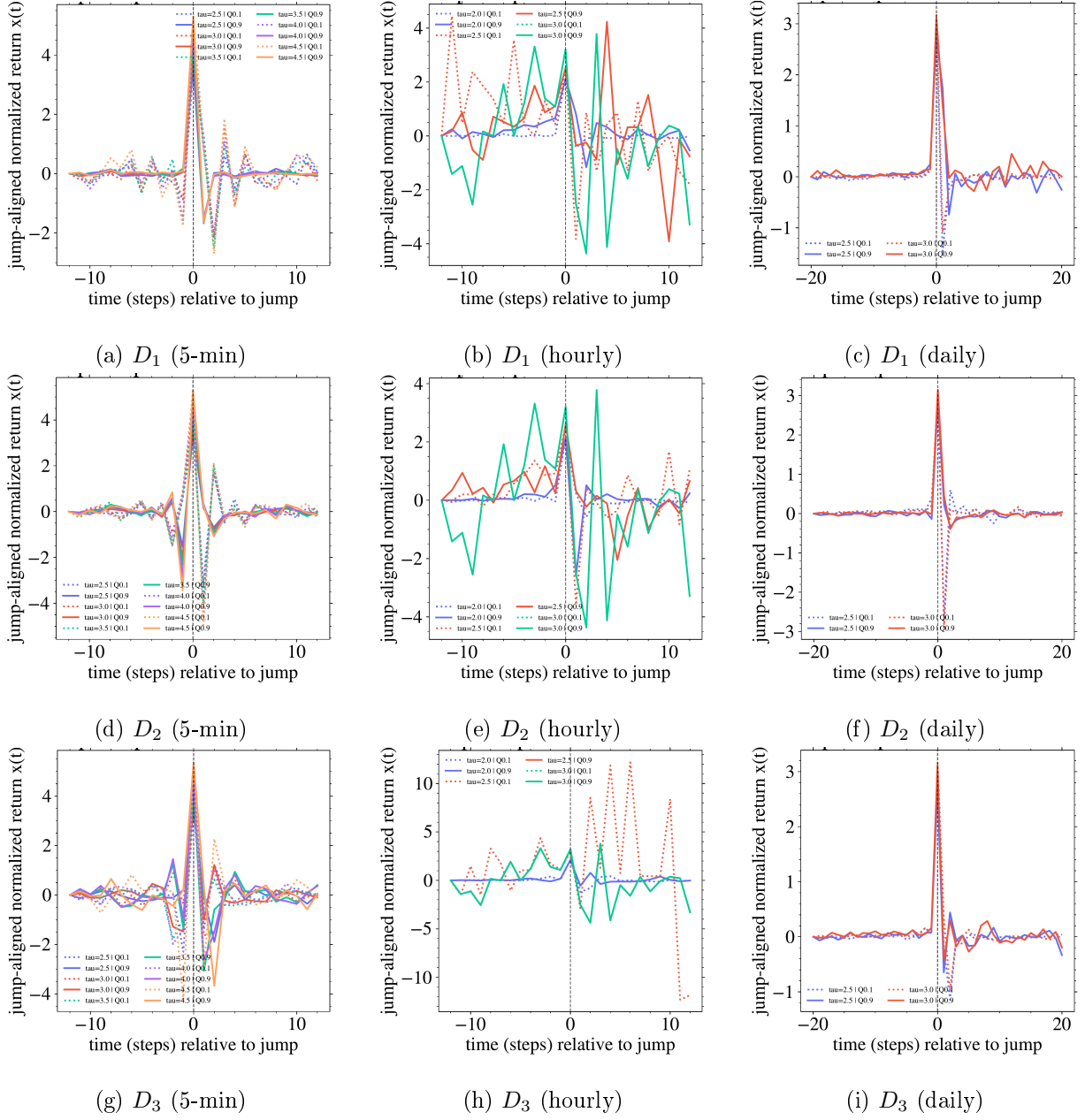


Figure 10: Threshold sweep overlays across frequencies (5-min, hourly, daily).